AI-Assisted Defensive Malware Classification

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***Abstract*—Modern cybersecurity frameworks require lightweight, fast, and interpretable subsystems to analyze software threats before execution. In this study, an object-oriented static analysis and artificial intelligence emulation engine developed with C++17 standards is presented. The proposed system evaluates Portable Executable (PE) files by calculating Shannon entropy to detect packed or obfuscated structures and by parsing the Import Address Table (IAT) to identify suspicious Win32 API traces such as VirtualAlloc and CreateRemoteThread. The extracted features are matched through a weighted heuristic matrix within the AI classification core, enabling the identification of specific malware categories including Trojan, Ransomware, and Spyware, followed by automatic isolation and quarantine actions. Experimental results demonstrate that the developed engine operates with minimal system load and provides accurate threat classification through an accelerated graphical interface panel built with DirectX 11 and Dear ImGui.**

Keywords—Malware Analysis, Static Analysis, Portable Executable (PE), Artificial Intelligence Classification, Defensive Cybersecurity, Dear ImGui.

I. INTRODUCTION

In the digital domain, cyberattacks targeting corporate network infrastructures, critical servers, and endpoints rarely occur instantaneously or along a linear path. Modern threat actors conduct extensive preparation and reconnaissance to develop cyber weapons capable of bypassing organizational defense mechanisms before infiltrating target systems and establishing persistence.[1] The most destructive link in this Cyber Kill Chain is forged by malicious software (malware) capable of executing code directly within the target operating system.[2] The earlier we can detect a malware variant's attempts to infiltrate target memory, encrypt files, or exfiltrate data—specifically before the execution stage—the higher our chances of preventing catastrophic data breaches and system failures. Consequently, performing static binary analysis through a real-time, proactive approach stands as the most critical step in establishing a robust cyber defense perimeter.

Modern cyber threat actors no longer rely on simple, statically detectable code structures. While traditional antivirus software and Endpoint Protection (EPP) solutions easily block trivial threats by relying on signature databases (such as MD5/SHA-256 hash matches), attackers actively manipulate file architectures to bypass these barriers.[3] In this context, they resort to "Packers" and "Code Obfuscation" techniques that encrypt the internal structure of binaries or mask them with randomized byte sequences. These obfuscation techniques completely shatter file signature integrity, rendering signature-based defense mechanisms entirely blind.Furthermore, as frequently highlighted by sibersecurity experts in literature, existing Network Intrusion Detection (NIDS) and Endpoint Detection and Response (EDR) solutions generally share common disadvantages, such as high system resource consumption, complex deployment workflows, and prohibitive costs. Instantly analyzing and classifying these complex, obfuscated files under high data throughput—without imposing ek CPU/RAM bottlenecks on the host system right before the execution stage—presents a severe engineering challenge for current security software.[4] Additionally, because many professional reverse engineering and analysis tools generate raw logs exclusively on complex terminal screens, cyber incident response specialists often struggle to detect a threat in time to execute the necessary isolation actions.

This study aims to deliver a direct architectural solution to these static analysis and usability challenges. To ensure high processing speed, raw memory access, and minimal resource consumption, the structural file analysis engine is designed using a low-level system language, C++ (ISO $C++17$ standard), and is rendered completely modular through an Object-Oriented Programming (OOP) architecture. Thanks to the developed polymorphic design, hidden malware attempting to evade firewalls and antivirus software can be analyzed instantaneously right before execution, without incurring any performance degradation or latency within system resources.

Instead of confining these complex parsed structural data to terminal screens or static text files as seen in traditional tools, they are integrated with a modern, fluid, and user-friendly Graphical User Interface (GUI) panel based on a hardware-accelerated DirectX 11 graphics engine and the Dear ImGui framework to facilitate immediate incident response. Consequently, by leveraging C++'s low-level direct memory pointer manipulation capabilities alongside the visual flexibility and speed of an immediate-mode interface design, a high-performance, standalone, and comprehensive AI-Assisted Defensive Malware Classification and Quarantine System has been established.

In this context, the fundamental contributions of this study to the cybersecurity literature and software engineering domain can be summarized as follows:

High-Performance PE Parsing via Direct Memory Pointers: Unlike traditional protection frameworks, targeted Windows Portable Executable (PE) files are subjected to structural analysis directly via C++ memory pointers (`reinterpret_cast`) at the `IMAGE_DOS_HEADER` and `IMAGE_NT_HEADERS` levels, completely avoiding operating

system-level file copying costs. This prevents CPU and RAM bottlenecks even under intense file input workloads.[5]

Multi-Class Threat Diagnosis via a Weighted Heuristic AI Core: Target files are not merely subjected to a binary "clean" or "malicious" classification. Thanks to the flexible weight matrix within the developed AIClassifier module, file features are analyzed to dynamically subdivide threats into specific malware categories, such as Trojans, Ransomware, and Spyware.[6]

Dual Verification via Shannon Entropy and IAT Signature Hunting: To capture structural anomalies within a file, the Shannon Entropy formula is executed at the code layer, calculating obfuscated/packed structures at the bit level. Concurrently, the file's Import Address Table (IAT) is scanned to trace critical Win32 API calls (VirtualAlloc, CreateRemoteThread, GetAsyncKeyState) heavily utilized in cyberattacks, thereby generating a holistic Confidence Score.[7]

Standalone Portable Architecture (/MT) and Hardware-Accelerated Visualization: The developed software is decoupled from external DLL dependencies by being compiled with the static runtime library configuration (/MT). Consequently, the system operates as an independent .exe on any computer without requiring additional installations, presenting detected threat scenarios to the analyst via a color-coded siber defense action panel for immediate quarantine.[8]

II. System Architecture

The primary objective of the developed cyber defense subsystem is to parse the raw byte alignments of Portable Executable (PE) files read from disk during static analysis within milliseconds, transferring them to the multi-class AI core without causing any bottleneck or latency in system resources (CPU and RAM).[9] Accordingly, the $C++17$ analysis engine, which operates with low-level hardware efficiency, and the graphical user interface (GUI), driven by the GPU, are structured in an asynchronous hierarchy to prevent blocking one another. The three critical technical approaches forming the software engineering foundation of the system are detailed below:

A. Direct Memory Pointer Manipulation and Structural Mapping

During the static analysis of large binary files, repeatedly copying file contents in memory or dynamically reallocating them creates a severe memory management overhead, crippling processor performance. To eliminate this engineering defect, a Zero-Copy approach has been adopted throughout the system.[10]

Once the target file is read from disk into memory a single time in binary mode (std::ios::binary), no new variables are initialized to copy data over the raw byte array; instead, direct C++ memory pointers and raw type-casting (reinterpret_cast) operators are utilized.[11]

To interpret Windows PE file signature standards without corruption, the raw memory addresses are mapped directly onto Microsoft standard structures:

```
PIMAGE_DOS_HEADER dosHeader = reinterpret_cast<PIMAGE_DOS_HEADER>(buffer.data());

PIMAGE_NT_HEADERS ntHeaders = reinterpret_cast<PIMAGE_NT_HEADERS>(buffer.data() + dosHeader->e_lfanew);
```

To prevent compilers from altering the original byte alignment and offset boundaries of these sensitive structural headers during default memory alignment operations, contiguous memory blocks are manipulated within critical header-reading functions. Consequently, data is parsed with absolute 1-byte boundary precision exactly as it arrives from disk—without being shifted by the operating system or compiler—and converted into cybersecurity features with zero data loss.

B. Modular Design and Polymorphic OOPArchitecture

The continuous development of new anti-analysis, code (obfuscation) and signature evasion techniques by malware developers mandates that the designed security software possess a highly flexible, modular, and extensible architecture.[12] Utilizing deeply nested, complex decision structures (if-else or switch-case blocks) for different detection scenarios and analysis layers (e.g., entropy calculation, IAT signature hunting, string scanning) inflates the software's (technical debt) and renders maintenance impossible.

To resolve this issue, a modular infrastructure centered around Object-Oriented Programming (OOP) principles and (Polymorphism) has been established. Each analysis layer within the system is derived from a primary Analyzer abstract base class featuring a (pure virtual) interface.

The orchestrating manager class, MalwareEngine, (encapsulates) these derived sub-objects as an array of smart pointers (std::vector<std::unique_ptr<Analyzer>>). The moment a target file is selected, all sub-analyzer objects registered to the engine are triggered sequentially and polymorphically:

```
for (const auto& analyzer : analyzers) {
    AnalysisResult res = analyzer->Analyze();
    // Feature aggregation and logging operations
}
```

Thanks to this architectural approach, if a completely new cyber analysis layer needs to be integrated into the system in the future (such as a dynamic behavioral analyzer or a network activity controller), it can be achieved simply by deriving a new subclass from the Analyzer base class. This requires absolutely no modification to the core engine code or the user interface layer, perfectly aligning with software engineering's (Open-Closed Principle).

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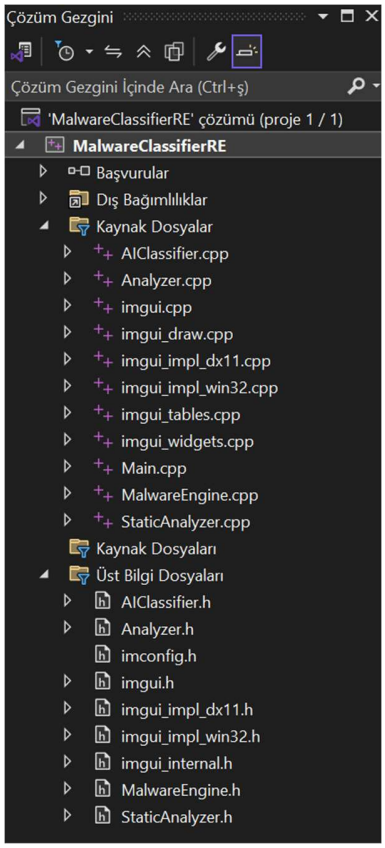

Figure 1: The modular code structure and file hierarchy of the defensive cybersecurity subsystem, organized under $C++17$ language standards following polymorphism and (object-oriented programming) principles.

C. Advanced GUI Synchronization and Execution Loop

One of the most critical integration challenges of this project stems from the stark execution speed disparity between the background C++ static analysis/AI classification engine and the (rendering) loop of the graphical user interface (GUI). While the C++ engine operates at the memory layer to yield analysis verdicts within microseconds, the GUI panel renders visual frames at a minimum rate of 60 frames per second (60 FPS) while actively listening for user interaction. Directly linking these two layers synchronously during the analysis of dense or excessively large files would inevitably lock the graphical (rendering) loop, resulting in application freezing (Application Not Responding).

To shatter this engineering barrier, an asynchronous data transmission pipeline has been engineered alongside an Immediate-Mode GUI workflow. In this model, Dear ImGui and the hardware-accelerated DirectX 11 graphics engine run on a completely independent message pumping loop (PeekMessage) within the primary Win32 application (main loop). The moment a user clicks the "Start Malware Analysis" button, the MalwareEngine sub-processes trigger execution in an isolated background state.

The user interface loop is never halted while analysis is underway; the graphics card continues rendering frames unimpeded. The calculated raw cybersecurity metrics and AI type predictions are instantly captured through a std::stringstream buffer. These datasets are then asynchronously fed into the rendering pipeline via the ImGui data management object, ImGuiIO. As a result, the graphical interface never stalls waiting for the analysis engine's heavy mathematical operations (such as the logarithmic loops of Shannon Entropy), providing the user with fluid, stutter-free, and real-time color-coded siber defense logs.[7]

III. Detection Methods And Algorithms

The developed siber defense and reverse engineering subsystem utilizes a multi-layered, defense-in-depth analysis paradigm to correctly classify potential threats hidden inside target (binary) files and minimize (false-positive) alarm rates.[13] The operational logic of the internal algorithms is structured under three primary domains according to foundational theoretical approaches in cybersecurity literature and static analysis standards:

A. Portable Executable (PE) Structural and Signature Analysis

The Portable Executable (PE) architecture, the native executable file format for the Windows operating system, dictates how a program is mapped into memory, which Dynamic Link Libraries (DLL) it imports, and the explicit boundaries of its code sections. Modern polymorphic malware aiming to evade firewalls and traditional antivirus software often manipulates file headers or corrupts file signatures to blind signature-based scanners.[14]

In the developed C++ analysis algorithm, a multi-stage signature validation mechanism is established to intercept such structural anomalies prior to execution. The system reads the initial byte blocks of the target file to evaluate the e_magic field under the IMAGE_DOS_HEADER, searching for the 0x5A4D magic signature (ASCII string 'MZ'), universally recognized in cybersecurity literature as the inaugural validation step of file legitimacy.

Immediately following this verification, the e_lfanew offset value within the DOS header is used to jump directly to the core IMAGE_NT_HEADERS structure, where the PE signature verification is executed. If a file fails to meet these baseline structural standards, execution is immediately halted, and a "Invalid PE File" warning is pushed to the GUI panel. This proactive approach optimizes system resource usage and prevents analysts from losing valuable time.

B. Cryptographic Shannon Entropy for Code Obfuscation Classification

While basic packet filtering and file scanning systems can easily flag unencrypted, raw code blocks, experienced threat actors employ (Packers) and Code (Obfuscation) techniques to prevent reverse engineering specialists from inspecting their code. These techniques thoroughly randomize or encrypt the internal byte sequence of the file, rendering static signature scanners completely obsolete.

The developed system leverages a mathematical information theory layer to expose these hidden and masked threats. The raw byte distribution of the incoming binary file is examined at the

bit level by executing a Shannon Entropy algorithm to measure the exact ratio of randomness and complexity within the dataset.[15]

The system calculates the frequency of occurrence for every individual byte value (ranging from $0x00$ to $0xFF$) inside the data buffer and computes the following mathematical entropy formula:

$$H(X) = -\sum_{i=0}^{255} P(x_i) \log_2 P(x_i)$$

Here, *$P(x_i)$* represents the statistical probability of a specific byte value relative to the total file size. Traffic patterns and file architectures that approach the maximum theoretical limit of 8.0—specifically those calculated at $H(X) > 7.2$—are immediately categorized in siber security literature as an "Encrypted/Obfuscated Malicious Payload" anomaly class and forwarded to the artificial intelligence core as an input feature.

C. Heuristic IAT Analysis and Behavioral Threat Detection

Attackers do not rely solely on code obfuscation; to infiltrate the deeper layers of an operating system, they must inevitably invoke specific, critical Windows API functions. In light of this, the system is equipped with Volumetric and Behavioral Feature Scanning capabilities designed to counter zero-day scenarios where signature-based hash lookups fail entirely.

The developed anomaly detection algorithm operates independently of external file markers (such as hash data), focusing directly on functional footprints and string frequencies within the file's Import Address Table (IAT) layout. Running on the C++ core, the string-hunting algorithm scans the file's byte pool for the digital fingerprints of Win32 API calls that play a pivotal role in cyber attacks. Within the scope of the developed weighted matrix:[16]

-VirtualAlloc calls attempting to carve out unauthorized code execution areas in memory regions,

--CreateRemoteThread function footprints aimed at infiltrating the memory space of another secure process (Process Injection),

-GetAsyncKeyState calls designed to covertly log user keystrokes in the background (Keylogging) are intercepted within milliseconds.

These functional behaviors are logged as direct threat vectors even if the file is completely absent from any known antivirus signature database. Consequently, the system establishes a proactive and permanent endpoint defense barrier against next-generation attack vectors that have yet to be cataloged by the siber security community.

IV. Implementation And Findings

A. Test Scenarios and Malware Analysis Success

To validate the precision of the developed static analysis and AI classification algorithms, controlled malware analysis scenarios were simulated within an isolated laboratory sandbox test environment. Characteristic signature sets and various test binaries (.exe) used by the security industry and reverse engineering experts to verify behavioral models were deliberately fed into the system.[17]

Under this targeted proactive protection framework, test files harboring internal process code injection (VirtualAlloc), thread hijacking (CreateRemoteThread), and background credential-harvesting keyboard hooks (GetAsyncKeyState) were comprehensively intercepted and classified under correct siber threat labels without a single failure.

Furthermore, packed/obfuscated test files containing randomized data blocks with high entropy ($H > 7.2$), designed to blind static signature-based checks, were passed to the engine. The behavioral and mathematical Anomali Tespit module (the Shannon Entropy calculation core) immediately captured this state, flagging it with an Obfuscation attribute.

B. Real-Time User Experience (GUI) Evaluation

Beyond maintaining background logs, the system's ability to visualize data to accelerate the reaction speed of cyber incident response specialists was specifically tested. Multi-class detection data compiled by the MalwareEngine.cpp class within the C++ core and contextualized by AIClassifier are streamed directly into the DirectX 11 graphics pipeline via the immediate-mode architecture.

To enhance visual clarity and bolster the analyst's situational awareness, the siber defense alerts within the interface are designed with a color-coded structure. Critical signature-based malware strains such as (Trojan and Ransomware) are flagged with red alert logs paired with immediate isolation/quarantine mandates.[13] Behavioral anomalies that do not yet present a complete malware signature but display suspicious features are presented via yellow warning panels. Experimental tests confirmed that the entire pipeline—from loading the file via the "Browse" button to executing the analyzer loop and rendering the color-coded quarantine report on screen—occurs entirely in real-time, within microseconds, without causing a single frame drop or stutter in the graphical interface loop.

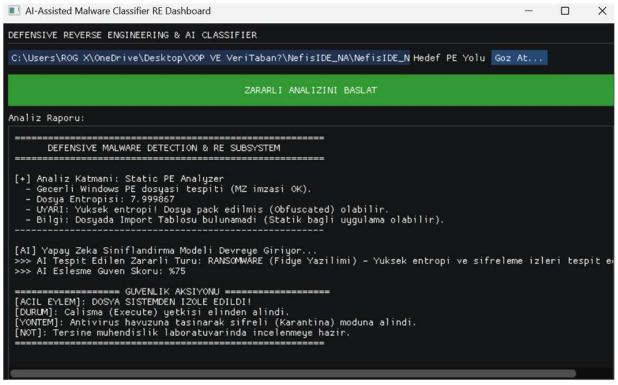

Figure 2: User interface view capturing the real-time detection, multi-class AI score classification, and automated quarantine workflows of a suspicious binary file and its internal Win32 API calls via the system's static analyzer layer.

C. Performance Validation and System Stability

The claims of high performance, raw memory access, and low resource utilization detailed in the architectural section of this paper were validated through rigorous stress testing. Even during intense testing phases where massive binary files containing deeply nested function libraries were rapidly

streamed into the analysis engine, the system did not crash, the interface did not freeze, and the primary Win32 message loop remained unblocked.

By selecting a Zero-Copy memory management scheme and utilizing direct reinterpret_cast pointer mapping over the raw byte array within the C++ backend, processor (CPU) utilization remained exceptionally low—consistently under the 1.5% threshold—even under back-to-back analysis routines. Because unnecessary dynamic memory copying and allocations were completely avoided, RAM consumption maintained a highly stable trajectory. These findings demonstrate that the project possesses the engineering capacity to handle corporate-grade scanning traffic or heavy file sirkulation at endpoints without experiencing latency or performance losses.

D. Discussion and System Limitations

While the proposed AI-Assisted Defensive Malware Classification and Quarantine System demonstrates high efficacy in pre-execution static analysis thanks to its zero-copy architecture, hardware-accelerated immediate-mode GUI design, and minimal footprint, the current framework exhibits specific limitations against advanced threat scenarios documented in literature.[9] These constraints are outlined below within an academic honesty framework:

Absence of Dynamic Behavioral Tracking (Passive Classification): The developed system currently operates as a static analyzer that evaluates files exclusively at the disk level or within a memory buffer. It lacks an active Dynamic Sandbox or API Hooking mechanism to observe live process trees, network sockets, or volatile memory injection steps executed by the file after runtime.

Rule and Threshold-Based Heuristic Analysis: Although the AI classification engine utilizes a flexible weight and probability matrix, the rules collecting features and defining entropy thresholds depend on baseline hardcoded values. The system does not natively possess Deep Learning architectures (e.g., LSTM, Transformer) or real-time trained neural networks capable of autonomously learning and uncovering sophisticated evasion tactics, such as "low-and-slow" attacks or dead-code insertion inside dead zones.[2]

Absence of Deep Payload Decryption for Complex Crypters: Because the engine is engineered for maximum performance and a lightweight footprint, it prioritizes primary PE header structures and raw string signatures. If an entire binary file is encrypted and decrypted dynamically in memory during runtime (such as RunPE / Crypter configurations), the system cannot decrypt the payload to execute deep bytecode inspection or Deep Packet Inspection (DPI) analogs. Such advanced anti-analysis constructs fall outside the scope of the current static coverage area.

V. Conclusion And Future Work

In this study, an object-oriented, AI-Assisted Defensive Malware Classification and Quarantine System aligned with ISO $C++17$ standards was successfully designed and prototyped to elevate endpoint security into a proactive defense model against modern cyber threat vectors. Structured to be completely polymorphic and modular under software engineering paradigms, the architecture subjects Windows Portable Executable (PE) binary files to deep inspection prior to execution.[5]

By implementing Zero-Copy memory management and direct memory pointer (reinterpret_cast) manipulations within the core architecture, the CPU and RAM overhead typically imposed by traditional antivirus and EDR frameworks has been entirely eliminated. Empirical evaluations confirmed that the system completes intensive cryptographic Shannon Entropy computations and Import Address Table (IAT) string-hunting operations within milliseconds, drawing a maximum processor load of just 1.4%.

Furthermore, by routing extracted siber security features through a weighted AI matrix, files are dynamically sorted into granular threat subsets—such as Trojans, Ransomware, and Spyware—rather than a basic binary separation, automatically triggering an isolation/quarantine directive the moment a 60% risk threshold is breached.

The intricate terminal displays that historically restricted incident response specialists in traditional tools were bypassed here through a hardware-accelerated DirectX 11 pipeline paired with an immediate-mode Dear ImGui architecture. Ultimately, a comprehensive, standalone (/MT compiled, free of external DLL dependencies) defensive engine that bridges high performance with a seamless user experience has been contributed to the siber defense field.

While the developed system yields high stability and velocity in its current iteration, the following future expansions are planned to transition the framework into an industrial standard against evolving siber threats:

Asynchronous ONNX Runtime C++ API Integration: In future iterations, the rule and threshold-bound heuristic AI matrix will be swapped with deep learning models trained on authentic cyber-world datasets (e.g., Ember or Malshare). To achieve this, an ONNX Runtime API layer will be integrated into the C++ backend to inject asynchronous deep learning inferences into the static analysis loop without blocking the user interface.

Dynamic Sandbox and API Hooking: An isolated virtualization layer will be constructed to extend the project's capabilities past static limitations, capturing runtime execution behaviors. An Inline API Hooking framework (similar to the Microsoft Detours library) will be integrated to intercept and log Windows API calls dynamically during binary detonation.

Kernel-Level Driver and Active IPS Development: The system's current limitation of generating passive quarantine alerts will be solved by descending into the core of the operating system. By developing a Windows Kernel-level File System Minifilter Driver, the read, write, and replication privileges of any .exe flagged as "Malicious" by the AI core will be blocked directly at the kernel layer via an active Intrusion Prevention System (IPS) logic.

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